

Exercises from chapter 3

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Exercise 1

First example:

$l(v) = 0$: it's trivially bounded, so it's continuous. So the Riesz representative is unique. Now we find it:

For the dot product:

$$(0, v) = \sum_{i=1}^n 0 \cdot v_i = 0 = l(v), \forall v \in \mathbb{R}^n \implies 0 \text{ is the representative}$$

For a matrix induced inner product: let A be a symmetric, positive-definite matrix. We define the inner product $(u, v)_A = u^\top A v$

$$(v, 0)_A = v^\top A 0 = v^\top 0 = 0 \implies 0 \text{ is the representative}$$

Second example:

$l_i(v) = v_i$: we need to prove it's bounded for each inner product

With dot product:

First we prove boundedness.

$$|l_i(v)|^2 = |v_i|^2 \leq \sum_{j=1}^n v_j^2 = \|v\|_2^2 \implies |l_i(v)| \leq \|v\|_2$$

Now we find the Riesz representant.

$$(u, v) = \sum_{j=1}^n u_j v_j = v_i, \forall v \in \mathbb{R}^n \implies u_j = 0, j \neq i$$

So $u = (0, \dots, 0, 1, 0, \dots, 0)$, where the i -th component receives the 1.

With matrix induced inner product:

For boundedness:

Let $\lambda_1 > 0$ be the smallest eigenvalue of A . Using the Rayleigh quotient

$$R(v) = \frac{v^\top Av}{v^\top v} \geq \lambda_1 \implies v^\top Av \geq \lambda_1 \|v\|_2^2 \geq \lambda_1 v_i^2 \implies v_i^2 \leq \frac{v^\top Av}{\lambda_1}$$

Then

$$|v_i| = |l_i(v)| \leq \frac{1}{\sqrt{\lambda_1}} \|v\|_A$$

The representant: let a_i be the i -th column of A . Then

$$(u, v)_A = u^\top Av = v_1 u^\top a_1 + \dots + v_n u^\top a_n = v_i, \forall v \in \mathbb{R}^n \implies u^\top a_j = 0, \forall j \neq i$$

This tells us that u is orthogonal to $\text{span}\{a_1, \dots, a_{i-1}, a_{i+1}, \dots, a_n\}$. Treating the term that remains: $v_i u^\top a_i = v_i \implies u^\top a_i = 1$.

So the Riesz representant is the vector u such that $u^\top a_i = 1$ and u is orthogonal to $\text{span}(\{a_1, \dots, a_n\} \setminus \{a_i\})$

Exercise 2

Let $V = \text{span}\{1, x, \dots, x_n\}$. Let's prove that V with the L^2 inner product is complete:

Note first that if $u \in V$, u is bounded (because $\Omega = (-1, 1)$ is bounded). Let $u_{max} = \sup_{x \in (-1, 1)} |u(x)|$. We have

$$\|u\|_{L^2(\Omega)} = \left(\int_{-1}^1 u^2(x) dx \right)^{\frac{1}{2}} \leq (2u_{max}^2)^{\frac{1}{2}} = \sqrt{2} |u_{max}| \quad (1)$$

This implies that if u_k is a Cauchy sequence in $\|\cdot\|_{L^\infty(\Omega)}$, then u_k is a Cauchy sequence in $\|\cdot\|_{L^2(\Omega)}$.

Take $u_k(x) = a_{0k} + a_{1k}x + \dots + a_{nk}x^n$ a Cauchy sequence in $\|\cdot\|_{L^\infty(\Omega)}$. Take $\varepsilon > 0$. In particular, as $\{1, x, \dots, x^n\}$ forms a base, we have that $\forall i = 1, \dots, n$, $\exists M > 0$ such that $m, k > M$ implies

$$|(a_{ik} - a_{im})x^i| \leq |a_{ik} - a_{im}| < \varepsilon$$

As $\mathbb{R}$ is complete $a_{ik} \rightarrow a_i$ as $k \rightarrow \infty$. So $u_k \rightarrow u = a_0 + a_1x + \dots + a_nx^n$. Now, because of 1, $u_k \rightarrow u$ with the L^2 norm. So V is a Hilbert space with the L^2 inner product.

First linear functional:

$l(v) = 0$. Clearly bounded

Representative: $u \equiv 0$. In fact

$$(0, v)_{L^2} = \int_{-1}^1 0v(x) dx = 0$$

Second linear functional:

$$l(v) = \int_{-1}^1 v(x) dx$$

Boundedness: By Cauchy-Schwarz inequality

$$\left(\int_{-1}^1 v(x) \cdot 1 dx \right)^2 \leq \int_{-1}^1 v^2(x) dx \cdot \int_{-1}^1 1^2 dx = 2 \int_{-1}^1 v^2(x) dx$$

Therefore

$$\int_{-1}^1 v(x) \leq \sqrt{2} \|v\|_{L^2(\Omega)}$$

Finding the representant:

$$(u, v)_{L^2} = \int_{-1}^1 u(x)v(x) dx = \int_{-1}^1 v(x) dx \implies u(x) = 1$$

$$\therefore u \equiv 1$$

Finding the Riesz representant of $l(v)=v(0)$

We need to find u_L such that, for all $v \in V$:

$$\int_{-1}^1 u_L(x)v(x) dx = v(0)$$

Remember first that the Legendre polynomials are orthogonal in the interval $(-1, 1)$. So, if $P_n(x)$ is the n -th Legendre polynomial:

$$\int_{-1}^1 P_n(x)P_m(x) dx = \frac{2}{2n+1} \delta_{mn}$$

So $P_1, \dots, P_n$ forms a base for this function space. Take the representations $u_L(x) = \sum_{k=0}^n u_k P_k(x)$ and $v(x) = \sum_{k=0}^n v_k P_k(x)$ in this base of Legendre polynomials. Then

$$\begin{aligned} \int_{-1}^1 u_L(x)v(x) dx &= \int_{-1}^1 \left(\sum_{k=0}^n u_k P_k(x) \right) \left(\sum_{i=0}^n v_i P_i(x) \right) dx = \\ &= \sum_{k=0}^n \sum_{i=0}^n u_k v_i \int_{-1}^1 P_k(x) P_i(x) dx \end{aligned}$$

Using the orthogonality:

$$\int_{-1}^1 u_L(x)v(x) dx = \sum_{k=0}^n u_k v_k \frac{2}{2k+1}$$

From another point of view,

$$v(0) = \sum_{k=0}^n v_k P_k(0) = \sum_{k=0}^n u_k v_k \frac{2}{2k+1}$$

Note that if

$$P_k(0) = u_k \frac{2}{2k+1} \implies u_k = \frac{2k+1}{2} P_k(0), \forall k = 0, \dots, n$$

We have a representant. As the Riesz representant is unique, we found u_L

Exercise 3

Let $u, v \in V$. By Cauchy-Schwarz inequality

$$(u, v) \leq \|u\|_V \|v\|_V$$

Where $\|\cdot\|_V$ is the induced norm. The bilinearity comes from the linearity in the first argument and the symmetry of the inner product. With this, a scalar product is a bilinear form.

Exercise 4

Symmetry: follow from the symmetry of the bilinear form.

Linearity: $a(\alpha v + \beta w, u) = \alpha a(v, u) + \beta a(w, u)$, by the bilinearity

Positivity: As the bilinear form is strongly coercive, $\exists \alpha > 0$ such that $a(u, u) \geq \alpha \|u\|_V^2 \geq 0$. Now remains to prove that $a(u, u) = 0 \iff u = 0$

$$(\implies) a(u, u) = 0. \text{ We have } a(u, u) = 0 \geq \alpha \|u\|_V^2 \implies \|u\|_V = 0 \implies u = 0$$

$$(\impliedby) u = 0 \implies a(u, u) = a(u + u, u) = 2a(u, u) \implies a(u, u) = 0$$

So a symmetric, continuous and coercive bilinear form defines an inner product.

Let $\{v_n\} \in V$ such that $\|v_n - v\|_V \xrightarrow{n \rightarrow \infty} 0, v \in V$. As a is continuous:

$$\|v_n - v\|_a = \sqrt{a(v_n - v, v_n - v)} \leq \sqrt{C_a} \|v_n - v\|_V \rightarrow 0 \implies \|v_n - v\|_a \xrightarrow{n \rightarrow \infty} 0$$